

NEXSTEP

GROUP NO: 08

GROUP MEMBERS:

1. HIMASAI VIHAR SURABOINA (2024BCS0225)
2. EDIGA VIVEK (2024BCS0229)
3. KURRI HIARANYA VENKATA REDDY (2024BCS0221)

1. Introduction

1.1 Purpose

NexStep is an AI-powered Competitive Exam Eligibility and Guidance System designed to help students identify competitive examinations they are eligible for based on their academic and personal profiles. The system provides personalized exam recommendations, eligibility analysis, application tracking, document management, and automated reminders. The purpose of this Software Requirements Specification (SRS) is to define the functional and non-functional requirements of the NexStep system.

1.2 Document Conventions

The following abbreviations are used throughout this document:

Term	Description
AI	Artificial Intelligence
SRS	Software Requirements Specification
UI	User Interface
Admin	System Administrator
Student	Registered User
OTP	One-Time Password
API	Application Programming Interface

1.3 Intended Audience and Reading Suggestions

This document is intended for:

- Project Guide
- Developers
- Testers
- Project Team Members
- End Users (Students)

Readers are encouraged to review the Introduction and Overall Description sections first, followed by the Functional and Non-Functional Requirements.

1.4 Product Scope

NexStep is a web-based application that assists students in discovering competitive examinations they are eligible for. It provides AI-powered eligibility analysis, personalized exam recommendations, document verification, application tracking, and automated notifications through a centralized platform. The system aims to simplify the exam discovery process, reduce manual effort, and ensure students do not miss important opportunities.

1.5 References

- IEEE Software Requirements Specification (SRS) Template
- Official Competitive Examination Notifications
- Official Examination Websites
- Project Requirement Document

2. Overall Description

2.1 Product Perspective

NexStep is a standalone web-based application designed to assist students in identifying competitive examinations based on their eligibility. The system serves as a centralized platform where students can explore exam opportunities, understand eligibility criteria, manage application timelines, and receive personalized guidance. It reduces the need to visit multiple websites by providing relevant information in a single platform.

2.2 Product Functions

The major functions of the system include:

- User Registration and Login
- Student Profile Management
- AI-Based Exam Eligibility Analysis
- Personalized Exam Recommendations
- AI Chat Assistant for Exam Queries
- Document Upload and Verification
- Application and Deadline Tracking
- Notification and Reminder Management
- Dashboard for Students
- Admin Management of Exams and Notifications

2.3 User Classes and Characteristics

Student

- Registers and logs into the system.
- Maintains a personal profile.
- Checks eligibility for competitive exams.
- Receives personalized recommendations.
- Uploads required documents.
- Tracks application progress.
- Receives reminders and notifications.

Administrator

- Manages exam information.
- Updates eligibility criteria.
- Verifies and maintains exam records.
- Sends announcements and notifications.
- Monitors system activities.

2.4 Use Case Diagram



2.5 Use Case List

UC ID	Use Case	Primary Actor
UC-1	Register	Student
UC-2	Login	Student/Admin
UC-3	Manage Student Profile	Student

UC ID	Use Case	Primary Actor
UC-4	Check Exam Eligibility	Student
UC-5	View Recommended Exams	Student
UC-6	Ask AI Assistant	Student
UC-7	Upload Documents	Student
UC-8	Track Applications	Student
UC-9	Receive Notifications	Student
UC-10	View Dashboard	Student
UC-11	Manage Exams	Administrator
UC-12	Manage Users	Administrator
UC-13	Manage Notifications	Administrator

2.6 Operating Environment

The system shall operate in the following environment:

- Web browsers (Google Chrome, Microsoft Edge, Mozilla Firefox)
- Desktop and Laptop Computers
- Android and iOS Mobile Browsers
- Internet Connectivity

2.7 Design and Implementation Constraints

- Users must have internet access to use the system.
- Exam information shall be maintained by the administrator.
- The system shall use only publicly available examination information.
- User data shall be protected according to privacy and security requirements.

2.8 User Documentation

The system shall provide:

- User Guide
- Administrator Guide

- [FAQ Section](#)
- [Help and Support Page](#)

2.9 Assumptions and Dependencies

- Students provide accurate academic and personal information.
- Examination authorities publish accurate and up-to-date notifications.
- Users have a stable internet connection.
- The system depends on the availability of official examination information.

3. External Interface Requirements

3.1 User Interfaces

The NexStep system shall provide the following user interfaces:

- Login Page
- Registration Page
- Student Dashboard
- Student Profile Management Page
- Exam Eligibility & Recommendation Page
- AI Chat Assistant Interface
- Document Upload Interface
- Application Tracking Dashboard
- Notification Center
- Administrator Dashboard

The interfaces shall be user-friendly, responsive, and easy to navigate.

3.2 Hardware Interfaces

The system shall support the following hardware devices:

- Desktop Computers
- Laptop Computers
- Smartphones
- Tablets

Users shall be able to access the system using any internet-enabled device.

3.3 Software Interfaces

The system shall interact with the following software components:

- Web Browser
- Database Management System
- Artificial Intelligence Service
- Email Notification Service

These interfaces enable the system to provide personalized recommendations, store user information, and send notifications.

3.4 Communication Interfaces

The system shall communicate through:

- HTTP/HTTPS Protocol
- Internet Connectivity
- Email Communication Services

These communication interfaces shall enable secure data exchange between users and the system.

4. Functional Requirements

4.1 User Authentication

Description

The system shall allow users to create an account and securely access the application.

Functional Requirements

- FR-1: The system shall allow students to register using their personal details.
- FR-2: The system shall authenticate registered users during login.
- FR-3: The system shall allow users to log out securely.
- FR-4: The system shall allow users to reset their password if forgotten.

4.2 Student Profile Management

Description

The system shall allow students to create and maintain their academic and personal profiles.

Functional Requirements

- FR-5: The system shall allow students to create their profile.
- FR-6: The system shall allow students to update their profile information.
- FR-7: The system shall store the student's academic details for eligibility analysis.

4.3 Exam Eligibility Analysis

Description

The system shall determine the competitive examinations a student is eligible for based on the information provided.

Functional Requirements

- FR-8: The system shall analyze the student's profile.
- FR-9: The system shall display eligible examinations.
- FR-10: The system shall explain the eligibility status for each examination.
- FR-11: The system shall indicate reasons for ineligibility whenever applicable.

4.4 Exam Recommendation

Description

The system shall recommend competitive examinations relevant to the student's profile and interests.

Functional Requirements

- FR-12: The system shall recommend suitable examinations.
- FR-13: The system shall display examination details.
- FR-14: The system shall allow students to save preferred examinations.

4.5 AI Chat Assistant

Description

The system shall provide an AI-powered assistant to answer examination-related queries.

Functional Requirements

- FR-15: The system shall allow students to ask questions about examinations.
- FR-16: The system shall provide responses related to eligibility, application procedures, required documents, and examination details.

4.6 Document Management

Description

The system shall allow students to upload and manage required documents.

Functional Requirements

- FR-17: The system shall allow document uploads.
- FR-18: The system shall verify document availability.
- FR-19: The system shall notify students about missing required documents.

4.7 Application Tracking

Description

The system shall help students monitor the status of their selected examinations.

Functional Requirements

- FR-20: The system shall allow students to track selected examinations.
- FR-21: The system shall display important examination dates.
- FR-22: The system shall display application status.

4.8 Notification Management

Description

The system shall notify students about important examination events.

Functional Requirements

- FR-23: The system shall send registration reminders.
- FR-24: The system shall notify students about examination updates.
- FR-25: The system shall send deadline reminders.

4.9 Administrator Management

Description

The administrator shall manage examination information and system data.

Functional Requirements

- FR-26: The administrator shall add new examinations.
- FR-27: The administrator shall update examination information.
- FR-28: The administrator shall delete outdated examination records.
- FR-29: The administrator shall manage notifications.
- FR-30: The administrator shall monitor registered users.

5. Non-Functional Requirements

5.1 Performance Requirements

- NFR-1: The system shall respond to user requests within an acceptable time under normal operating conditions.
- NFR-2: The system shall support multiple users accessing the application simultaneously.
- NFR-3: The system shall efficiently process eligibility analysis and display results.

5.2 Safety Requirements

- NFR-4: The system shall prevent unauthorized modification of user information.
- NFR-5: The system shall protect stored data from accidental loss.
- NFR-6: The system shall ensure data consistency during system operations.

5.3 Security Requirements

- NFR-7: The system shall authenticate users before granting access.
- NFR-8: The system shall ensure that only authorized administrators can manage examination data.
- NFR-9: The system shall protect user information from unauthorized access.
- NFR-10: The system shall maintain the confidentiality of user data.

5.4 Software Quality Attributes

Availability

- The system shall be available whenever users require access, subject to scheduled maintenance.

Reliability

- The system shall provide accurate and consistent examination recommendations.

Usability

- The system shall provide a simple, user-friendly, and intuitive interface.

Maintainability

- The system shall support easy updates to examination information and eligibility criteria.

Scalability

- The system shall support the addition of new examinations and an increasing number of users without affecting functionality.

5.5 Business Rules

- BR-1: A student shall register before accessing personalized features.
- BR-2: Eligibility recommendations shall be based on the student's profile information.
- BR-3: Only administrators shall modify examination details.
- BR-4: The system shall notify students of important examination updates and deadlines.
- BR-5: Students shall be able to update their profile information whenever required.

5.6 Other Requirements

- The system shall support future addition of new competitive examinations.
- The system shall provide centralized access to examination-related information.
- The system shall maintain records of registered users and their selected examinations.

6. Other Requirements

6.1 Business Rules

- Only registered users shall access personalized services.
- Examination recommendations shall be generated based on the student's profile.
- Only administrators shall add, update, or remove examination information.
- Students shall receive notifications only for examinations they choose to track.
- Examination information shall be updated whenever official notifications are released.

6.2 Assumptions and Dependencies

- Students provide accurate academic and personal information.
- Examination authorities publish authentic and updated information.
- Users have access to a stable internet connection.
- The system depends on the availability of official examination notifications.

Appendix A: Glossary

Term	Description
AI	Artificial Intelligence
Eligibility	The conditions required to apply for a competitive examination
Dashboard	The main interface displaying user information and exam details
Administrator	User responsible for managing examination data
Student	Registered user of the system
Notification	Alert sent to users regarding important examination updates
Recommendation	Suggested examinations based on the student's profile

Appendix B: Acronyms and Abbreviations

Acronym	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
SRS	Software Requirements Specification
UI	User Interface
OTP	One-Time Password
HTTP	Hypertext Transfer Protocol

Acronym	Meaning
HTTPS	Hypertext Transfer Protocol Secure

UI WIREFRAMES

```

+-----+
|               NexStep               |
|                                     |
|  Email:      _____             |
|  Password:   _____             |
|                                     |
|      [ Login ]                      |
|                                     |
|  Don't have an account?             |
|      Register                       |
|                                     |
+-----+

```

```

+-----+
|               Register               |
|                                     |
|  Name:       _____             |
|  Email:      _____             |
|  Password:   _____             |
|  Degree:     _____             |
|  Year:       _____             |
|                                     |
|      [ Register ]                   |
|                                     |
+-----+

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+-----+
| NexStep Dashboard                   |
+-----+
| Welcome, Student                   |
|                                     |
| [Profile] [Eligible Exams] [AI Chat] |
| [Applications] [Notifications]      |
|                                     |
| Upcoming Deadlines                 |
| -----                           |
| GATE - Aug 20                      |
| CAT  - Sep 15                      |
|                                     |
+-----+

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Eligible Exams	
✓ GATE	Eligible
✓ CAT	Eligible
✗ UPSC	Not Eligible
[View Details]	

AI Assistant

Ask your question...

[Send]

AI: You are eligible for GATE because...

My Applications		
Exam	Status	Deadline
GATE	Applied	Aug 20
CAT	Pending	Sep 15

Admin Dashboard

[Manage Exams]

[Manage Users]

[Send Notifications]

[Update Eligibility Rules]